

Python 3

07	1	<u>Example Solution</u> <pre> text = input("Enter the text to compress: ") print ("The compressed text is: ", end="") LastChar = "" CountOfLastChar = 0 for Count in range(0, len(text)): if text[Count] == LastChar: CountOfLastChar += 1 else: if LastChar != "": print (LastChar, " " , CountOfLastChar, " ",end="") LastChar = text[Count] CountOfLastChar = 1 print (LastChar, " " , CountOfLastChar, " ") </pre>	12
09	1	<pre> def __InputCoordinate(self, CoordinateName): Coordinate = int(input(" Input " + CoordinateName + " coordinate:")) while Coordinate < 0 or Coordinate >= self.__LandscapeSize: Coordinate = int(input("Coordinate is outside of landscape, please try again.")) return Coordinate </pre>	4
10	1	<pre> def CalculateNewAge(self): super(Rabbit, self).CalculateNewAge() if self.__Gender == Genders.Male: self._ProbabilityOfDeathOtherCauses = self._ProbabilityOfDeathOtherCauses * 1.5 else: if self._Age >= 2: self._ProbabilityOfDeathOtherCauses = self._ProbabilityOfDeathOtherCauses + 0.05 </pre>	5
11	1	<pre> class Location: def __init__(self, TerrainType): self.Fox = None self.Warren = None self.Terrain = TerrainType </pre>	3
11	2	<pre> def __CreateLandscapeAndAnimals(self, InitialWarrenCount, InitialFoxCount, FixedInitialLocations): for x in range (0, self.__LandscapeSize): for y in range (0, self.__LandscapeSize): if x == 5 or y == 2: self.__Landscape[x][y] = Location("R") else: </pre>	3

		<pre> self.__Landscape[x][y] = Location("L") if FixedInitialLocations: ... </pre>	
11	3	<pre> def __DrawLandscape(self): print() print("TIME PERIOD:", self.__TimePeriod) print() print(" ", end = "") for x in range (0, self.__LandscapeSize): print(" ", end = "") if x < 10: print(" ", end = "") print(x, " ", end = "") print() for x in range (0, self.__LandscapeSize * 5 + 3): #CHANGE print("-", end = "") print() for y in range (0, self.__LandscapeSize): if y < 10: print(" ", end = "") print("", y, " ", sep = "", end = "") for x in range (0, self.__LandscapeSize): if not self.__Landscape[x][y].Warren is None: if self.__Landscape[x][y].Warren.GetRabbitCount() < 10: print(" ", end = "") print(self.__Landscape[x][y].Warren.GetRabbitCount(), end = "") else: print(" ", end = "") if not self.__Landscape[x][y].Fox is None: print("F", end = "") else: print(" ", end = "") print(self.__Landscape[x][y].Terrain, end = "") print(" ", end = "") print() </pre>	2
11	4	<pre> def __CreateNewWarren(self): x = random.randint(0, self.__LandscapeSize - 1) y = random.randint(0, self.__LandscapeSize - 1) while not self.__Landscape[x][y].Warren is None or self.__Landscape[x][y].Terrain == "R": x = random.randint(0, self.__LandscapeSize - 1) y = random.randint(0, self.__LandscapeSize - 1) if self.__ShowDetail: print("New Warren at (" , x, ", ", y, ")", sep = "") self.__Landscape[x][y].Warren = Warren(self.__Variability) self.__WarrenCount += 1 def __CreateNewFox(self): </pre>	3

		<pre> x = random.randint(0, self.__LandscapeSize - 1) y = random.randint(0, self.__LandscapeSize - 1) while not self.__Landscape[x][y].Fox is None or self.__Landscape[x][y].Terrain == "R": x = random.randint(0, self.__LandscapeSize - 1) y = random.randint(0, self.__LandscapeSize - 1) if self.__ShowDetail: print(" New Fox at (", x, ",", y, ")", sep = "") self.__Landscape[x][y].Fox = Fox(self.__Variability) self.__FoxCount += 1 </pre>	
12	1	<pre> def CheckIfPathCrossesRiver(self, FoxX, FoxY, WarrenX, WarrenY): if FoxX - WarrenX > 0: xChange = 1 else: xChange = -1 if WarrenX != FoxX: x = WarrenX + xChange while x != FoxX: if self.__Landscape[x][FoxY].Terrain == "R": return True x += xChange if FoxY - WarrenY > 0: yChange = 1 else: yChange = -1 if WarrenY != FoxY: y = WarrenY + yChange while y != FoxY: if self.__Landscape[FoxX][y].Terrain == "R": return True y += yChange return False </pre>	9
12	2	<pre> def __FoxesEatRabbitsInWarren(self, WarrenX, WarrenY): RabbitCountAtStartOfPeriod = self.__Landscape[WarrenX][WarrenY].Warren.GetRabbitCount() for FoxX in range(0, self.__LandscapeSize): for FoxY in range (0, self.__LandscapeSize): if not self.__Landscape[FoxX][FoxY].Fox is None: if not self.CheckIfPathCrossesRiver(FoxX, FoxY, WarrenX, WarrenY): #INDENTATION CHANGED AFTER THIS LINE Dist = self.__DistanceBetween(FoxX, FoxY, WarrenX, WarrenY) if Dist <= 3.5: PercentToEat = 20 elif Dist <= 7: PercentToEat = 10 else: PercentToEat = 0 RabbitsToEat = int(round(float(PercentToEat * RabbitCountAtStartOfPeriod / 100))) </pre>	2

	<pre>FoodConsumed = self.__Landscape[WarrenX][WarrenY].Warren.EatRabbits(RabbitsToEat) self.__Landscape[FoxX][FoxY].Fox.GiveFood(FoodConsumed) if self.__ShowDetail: print(" ", FoodConsumed, " rabbits eaten by fox at (", FoxX, ",", FoxY, ").", sep = "")</pre>	
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